Module 1: The foundation

Cloud setup & access

Cloud setup & access

Prerequisites

- **Goal:** A secure environment before deploying code.
- **Providers:** DigitalOcean, AWS EC2, Linode, or Raspberry Pi.

Cloud setup & access

Prerequisites

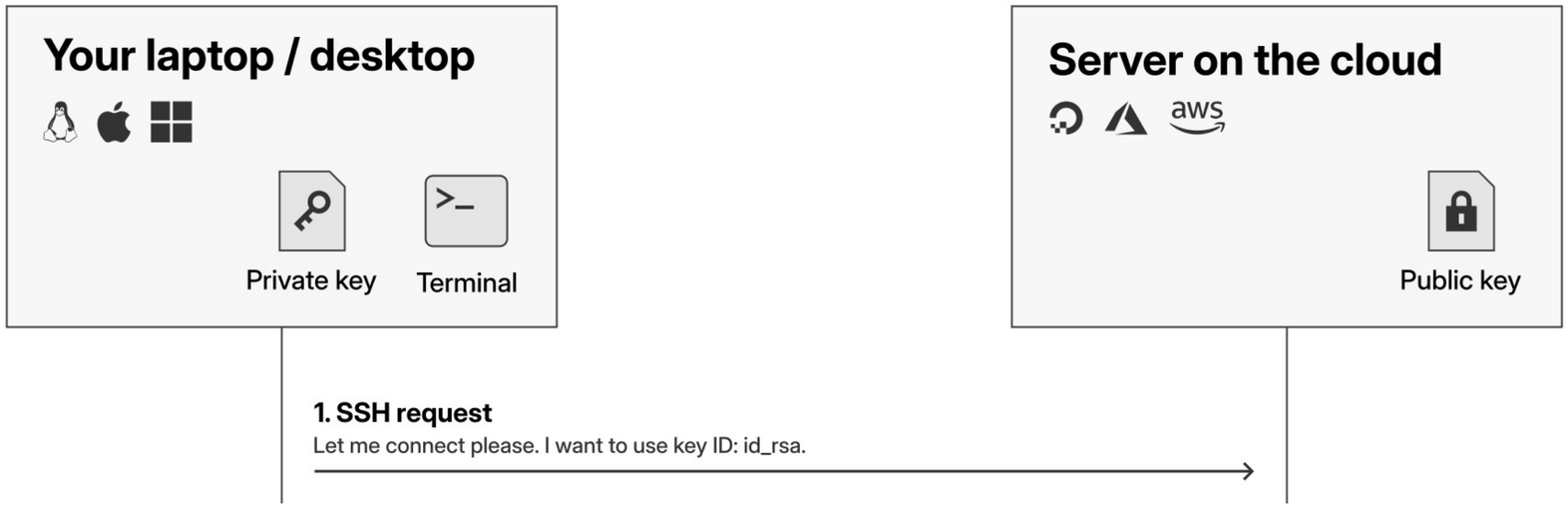
- **Goal:** A secure environment before deploying code.
- **Providers:** DigitalOcean, AWS EC2, Linode, or Raspberry Pi.

Learning objectives

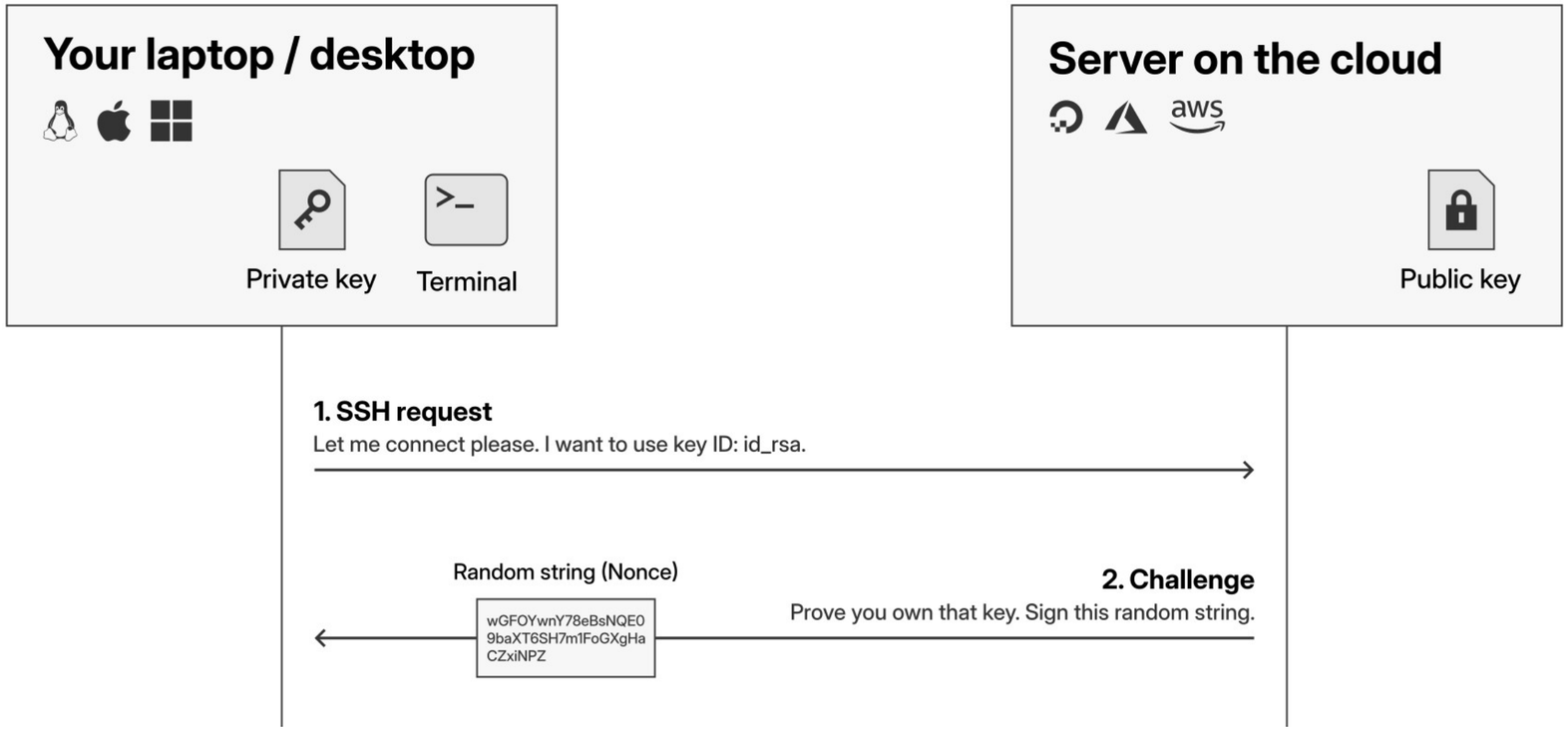
- Deploy a server running **Ubuntu 24.04 LTS**.
- Generate and configure **SSH Keys** (Ed25519).
- Understand the **"Lock & Key"** security model.
- **Security:** Disable password authentication completely.

How SSH keys work

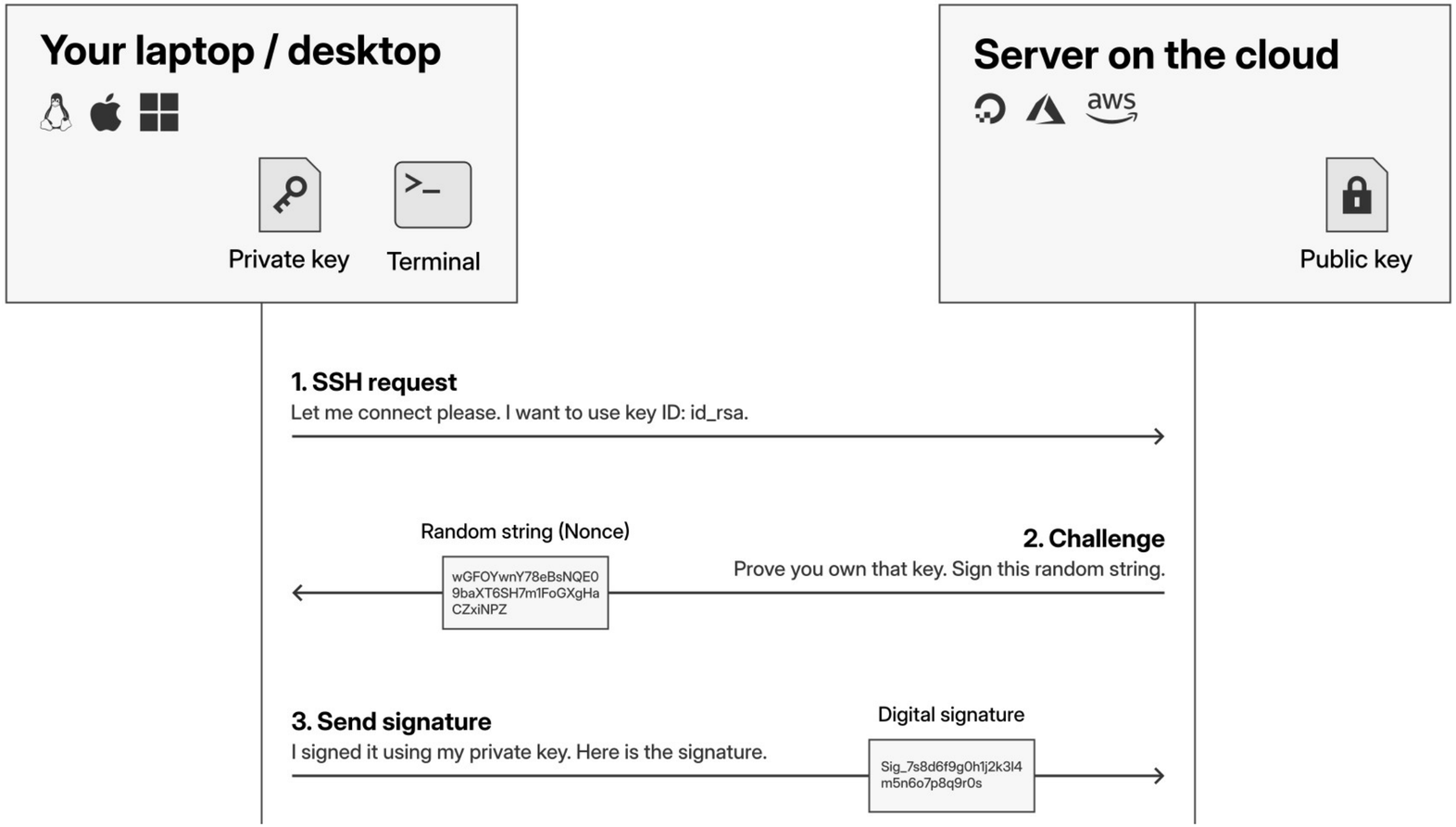
How SSH keys work



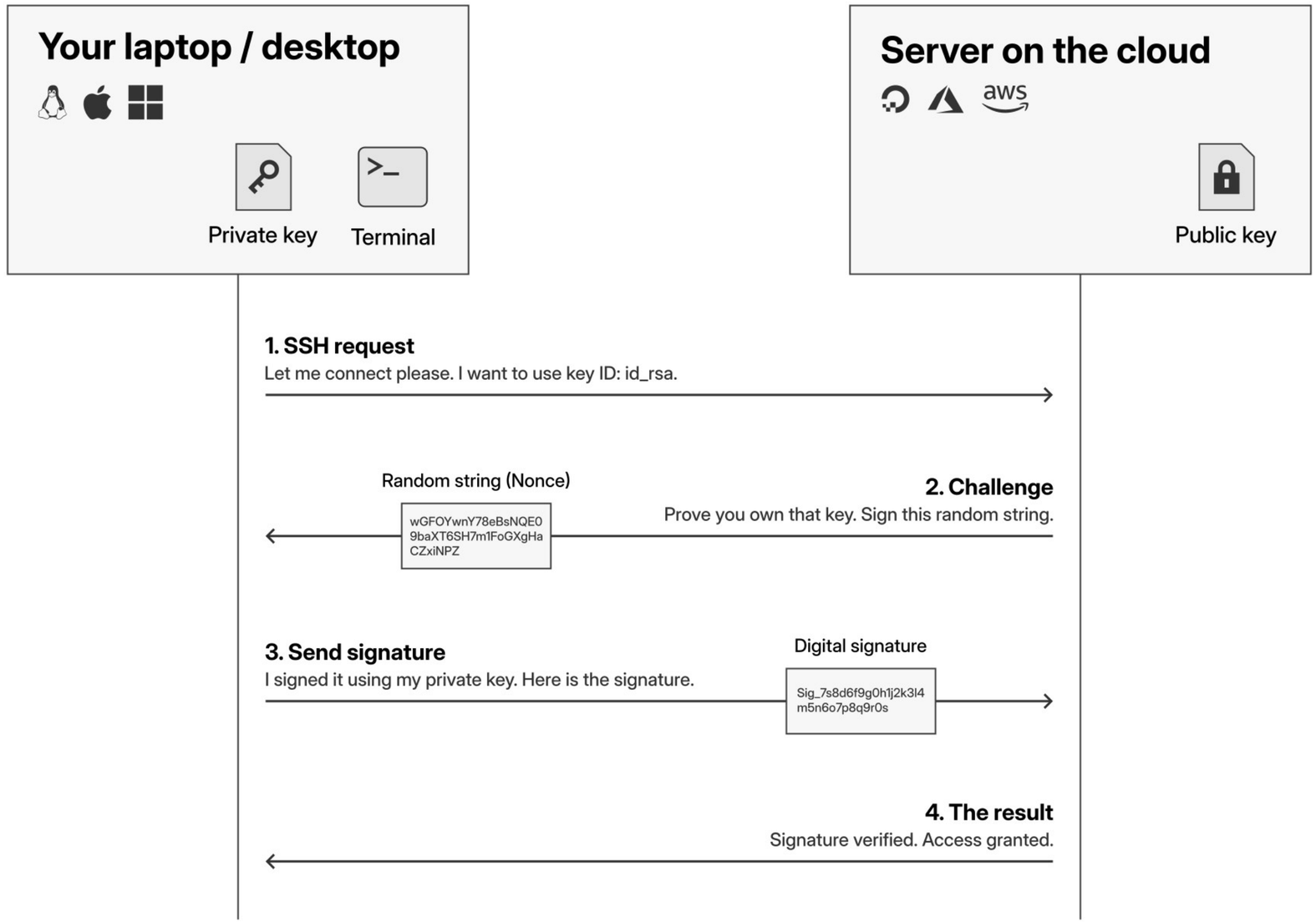
How SSH keys work



How SSH keys work



How SSH keys work



What is next?

- **Current status:** You now have a clean, secure Ubuntu server.
- **The risk:** The server is still exposed to the open internet.
- **Next steps:**
  - **Firewall configuration:** Setting up UFW (Uncomplicated Firewall).
  - **Brute force protection:** Installing Fail2Ban to block attackers.
  - **Emergency access:** How to use the recovery console.

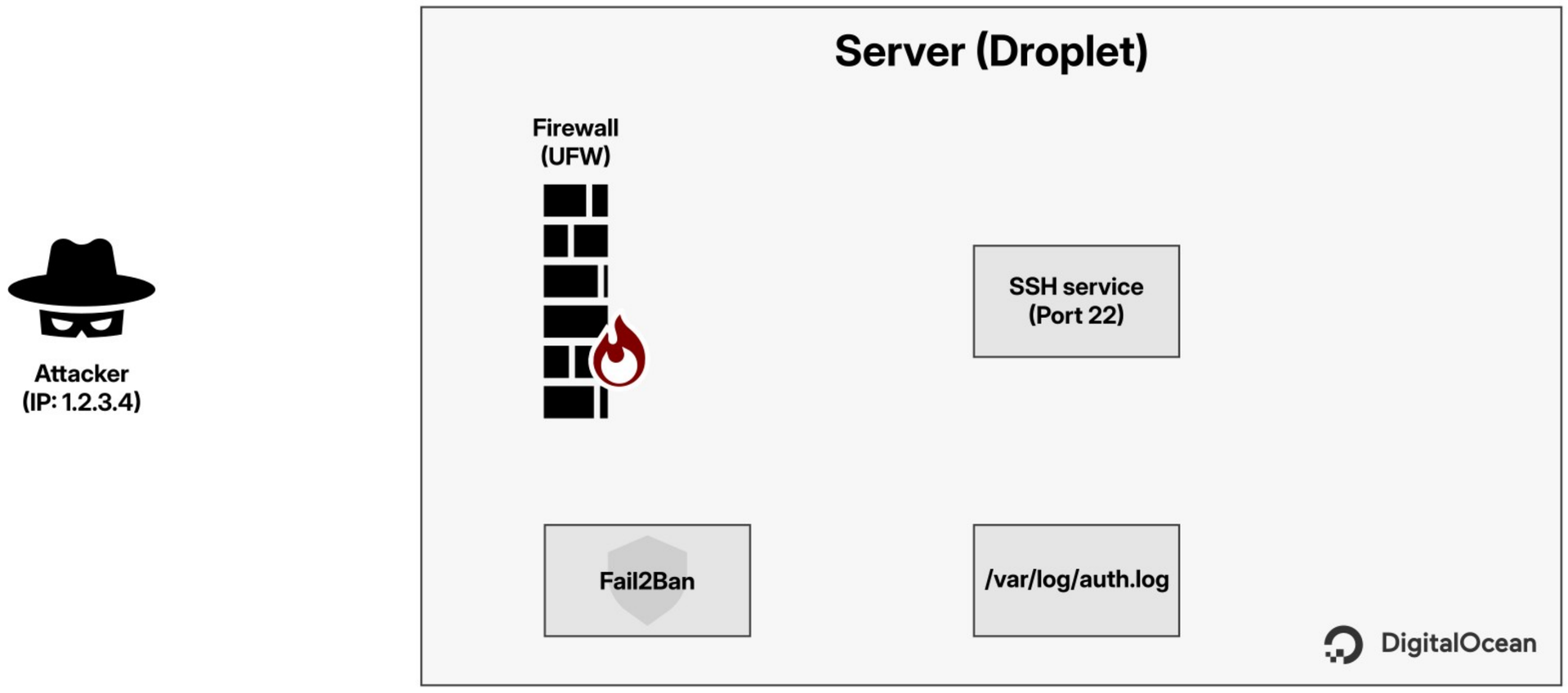
Module 1: The foundation

The firewall strategy

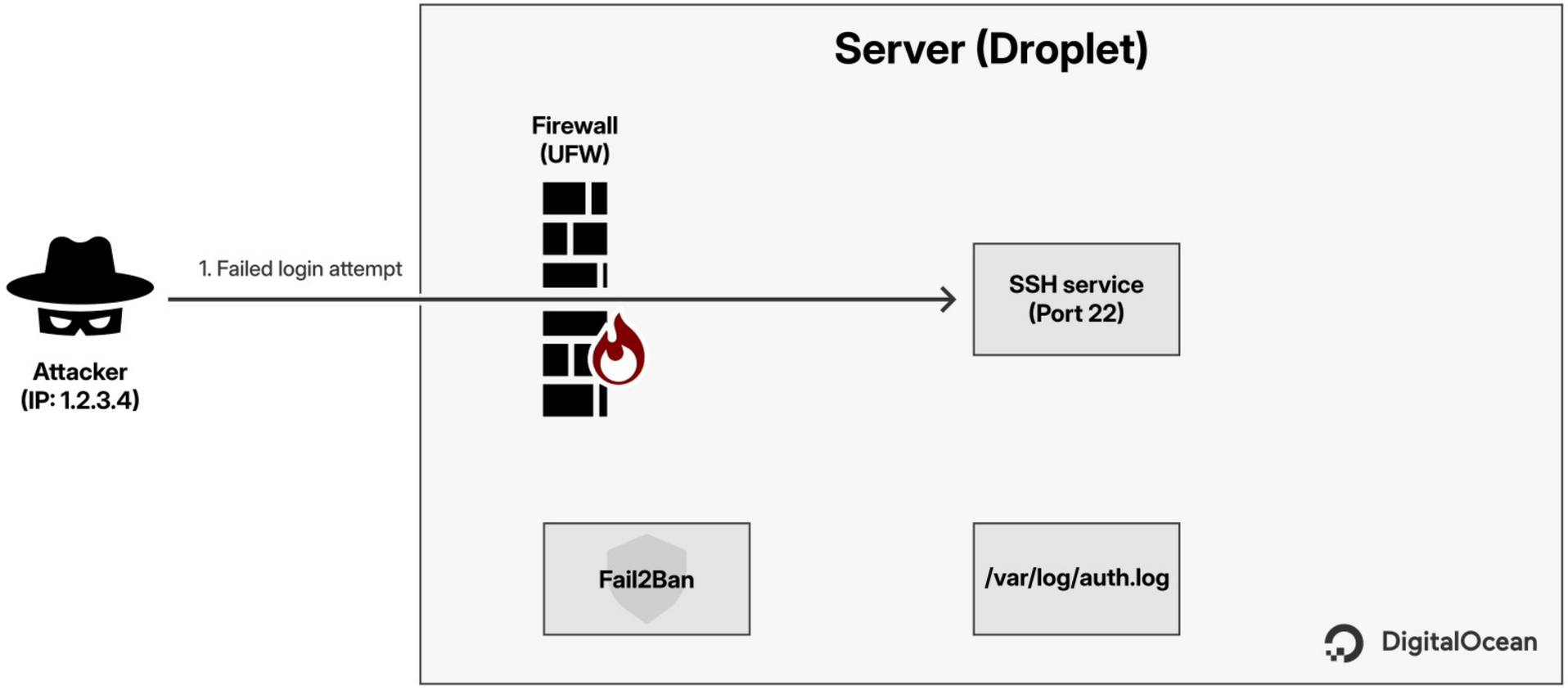
The firewall strategy

- **Problem:** Default Linux servers accept traffic on all ports.
- **Solution:** Use UFW to block everything by default.
- **Automation:** Use Fail2Ban to block bots instantly.
- **Safety net:** Using the DigitalOcean Recovery Console.

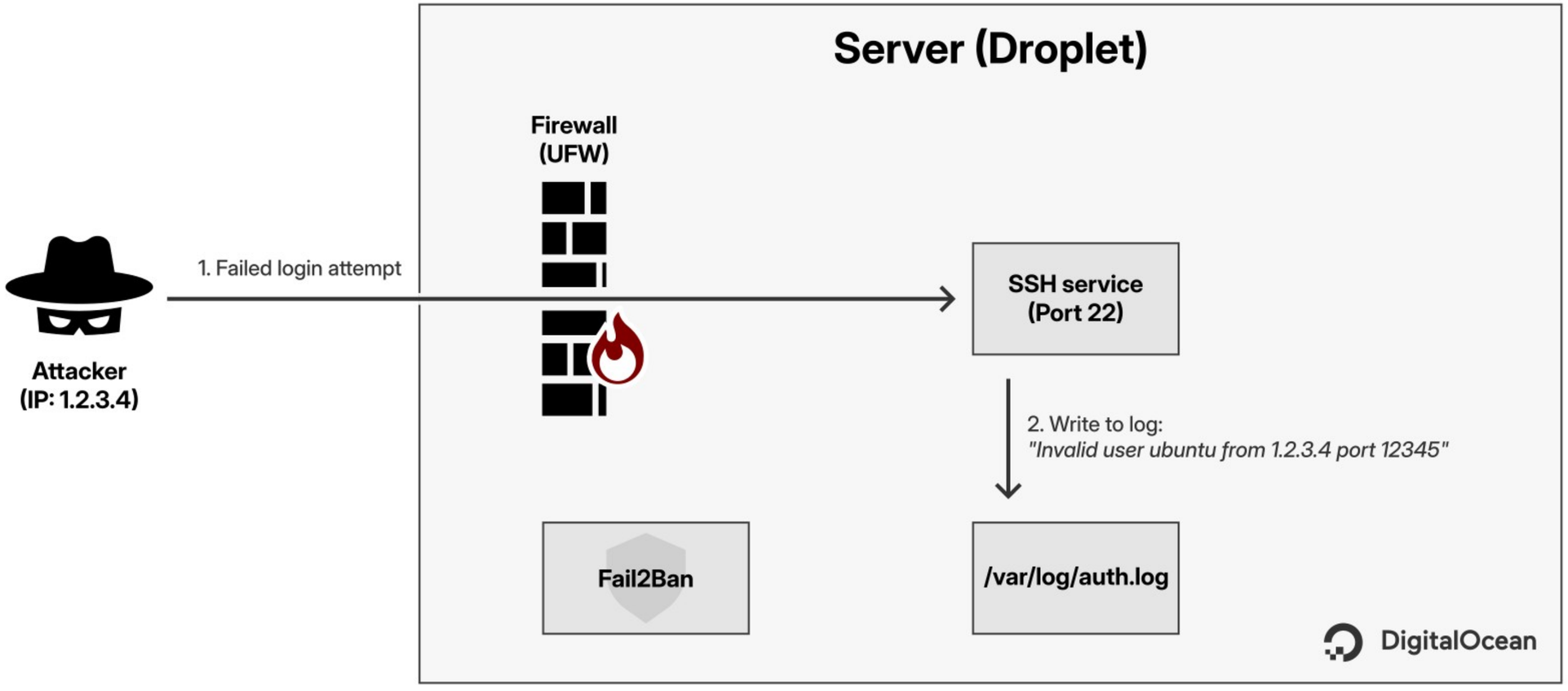
How Fail2Ban works



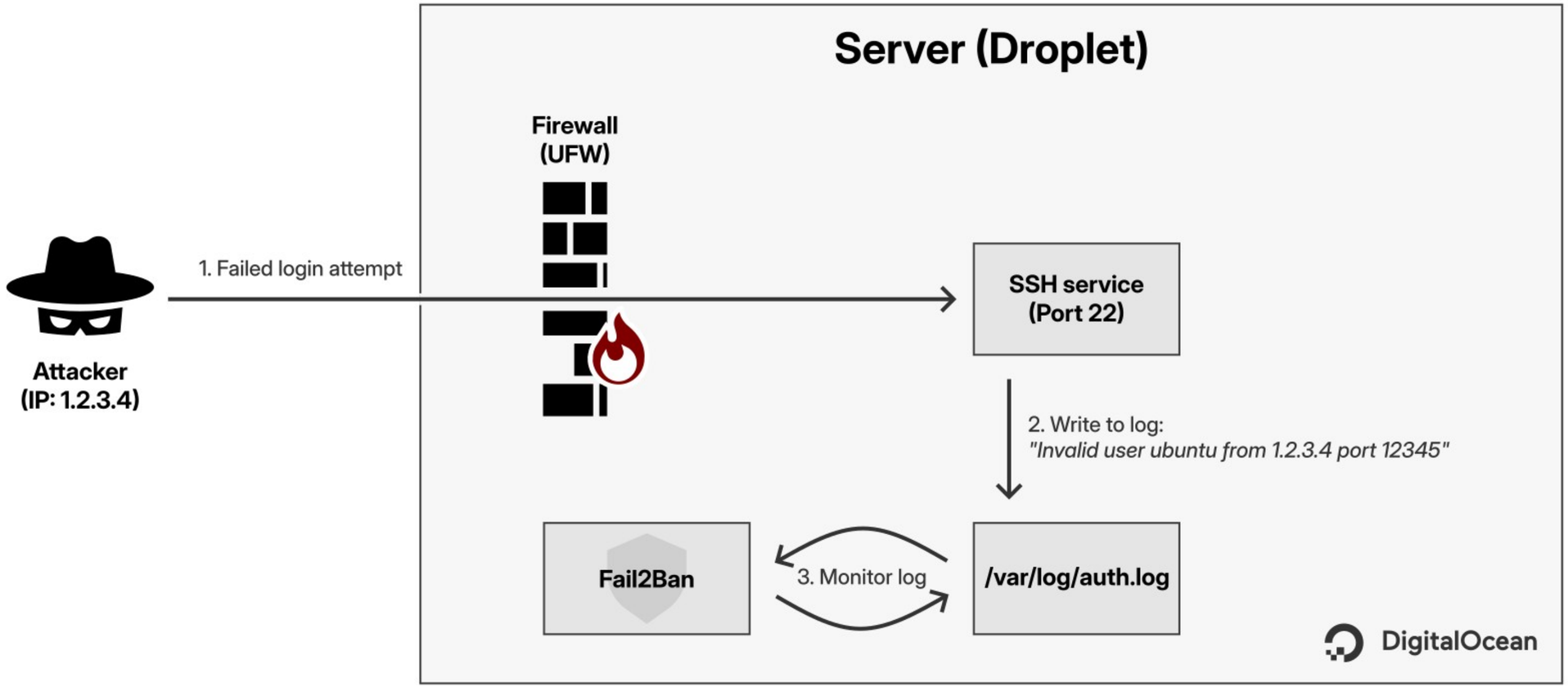
How Fail2Ban works



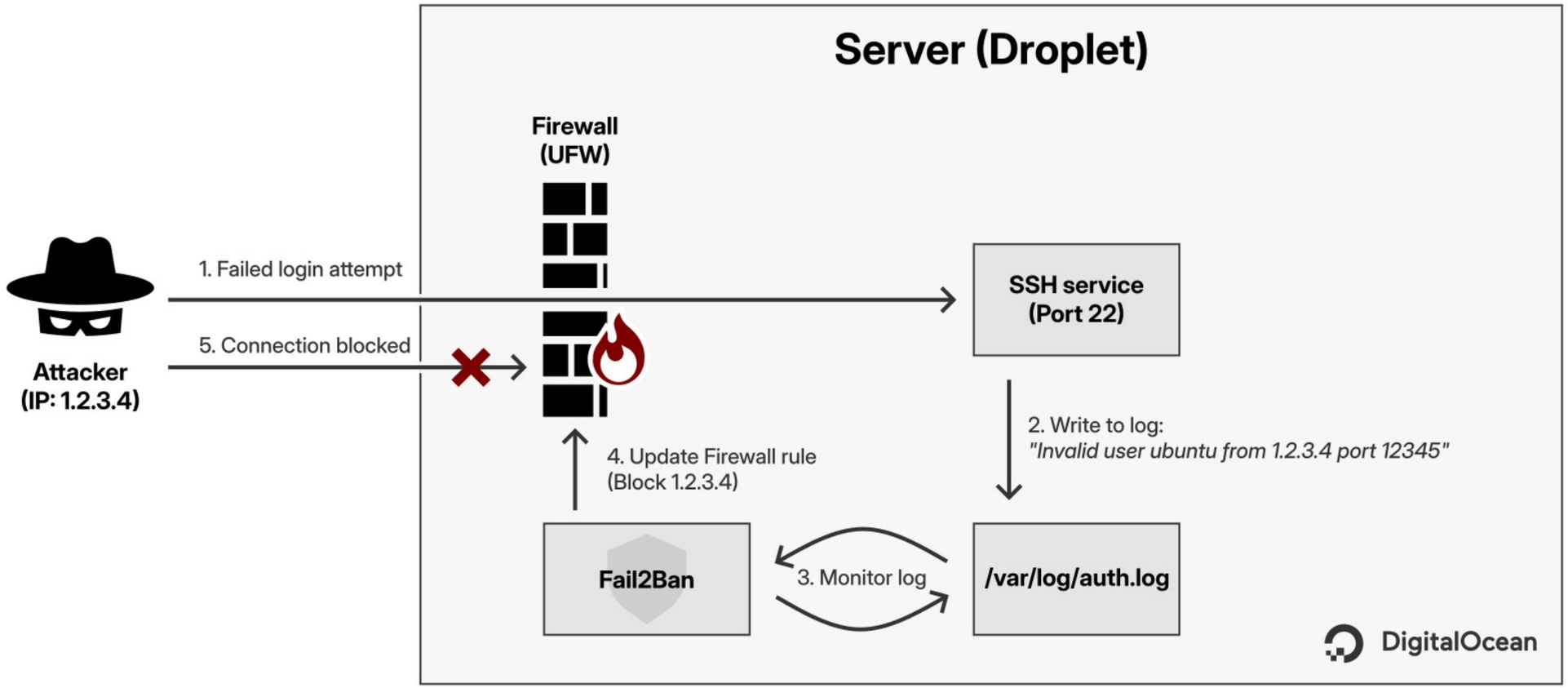
How Fail2Ban works



How Fail2Ban works



How Fail2Ban works

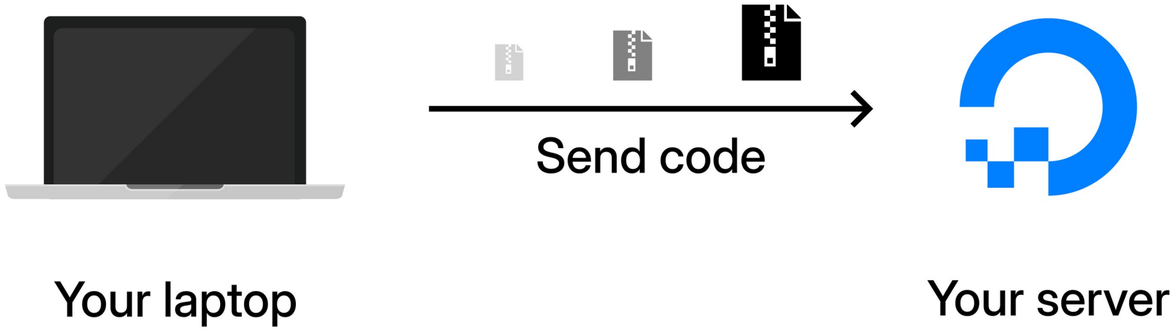


Module 2: The application runtime

Manual deployment

Manual deployment

- Move code using **scp** and organize permissions.

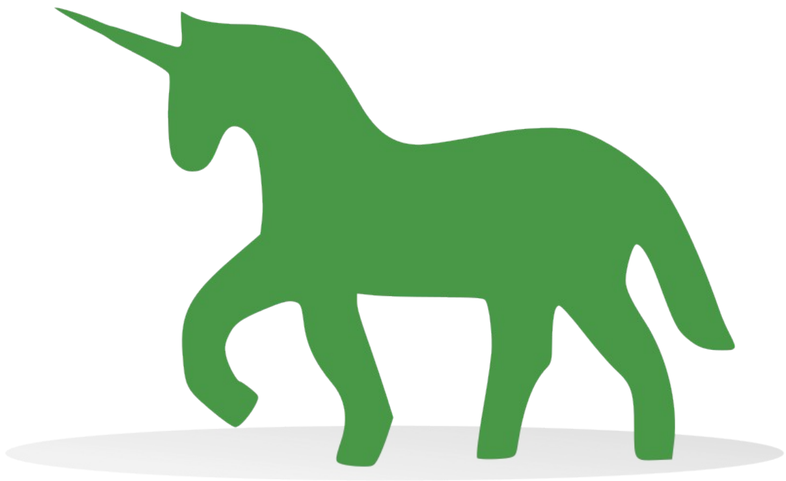


Manual deployment

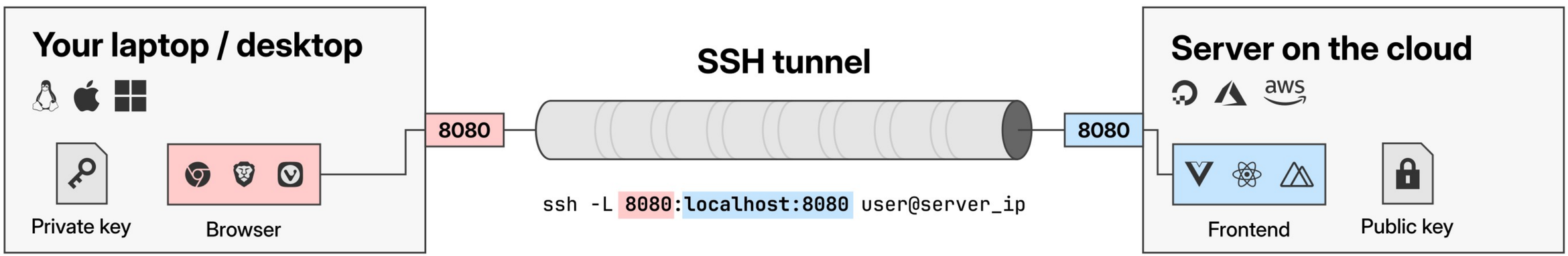
- Move code using **scp** and organize permissions.
- Install **Python** (Backend) and **Node.js** (Frontend).
- Fix build errors using **Swap files**.
- Use **SSH Tunnels** to bypass the **firewall**.

Manual deployment

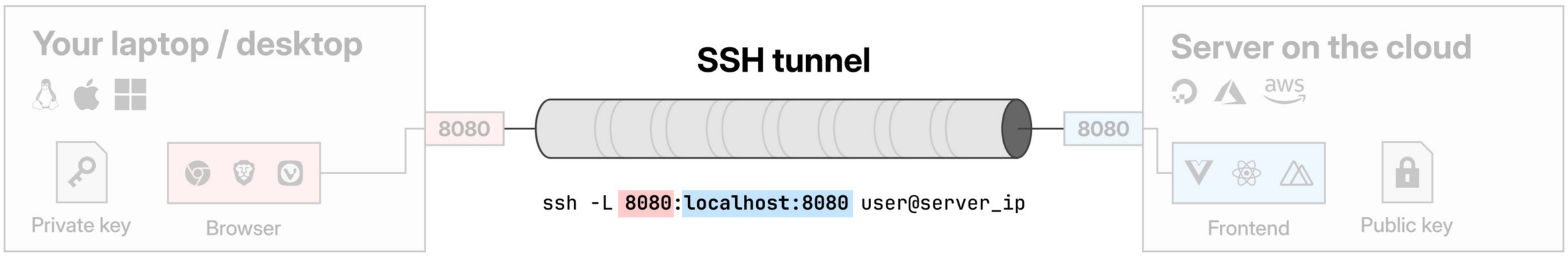
- Move code using **scp** and organize permissions.
- Install **Python** (Backend) and **Node.js** (Frontend).
- Fix build errors using **Swap files**.
- Use **SSH Tunnels** to bypass the **firewall**.
- Configure **Gunicorn** and **Supervisor**.



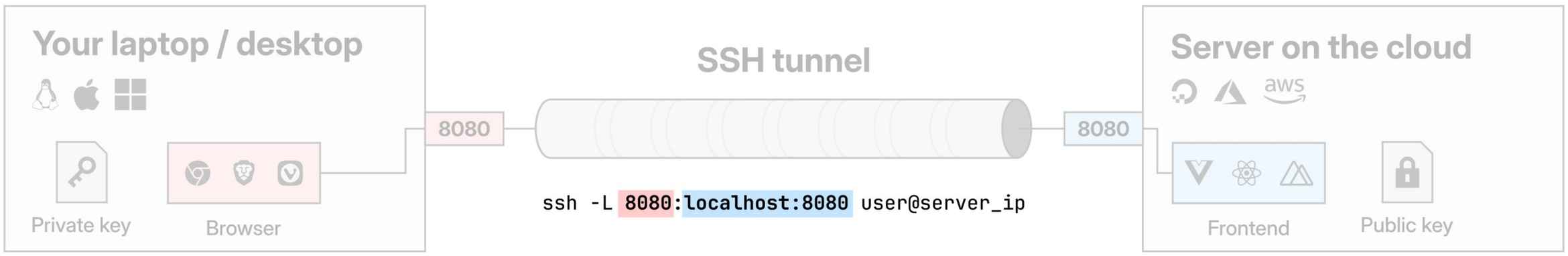
SSH tunneling



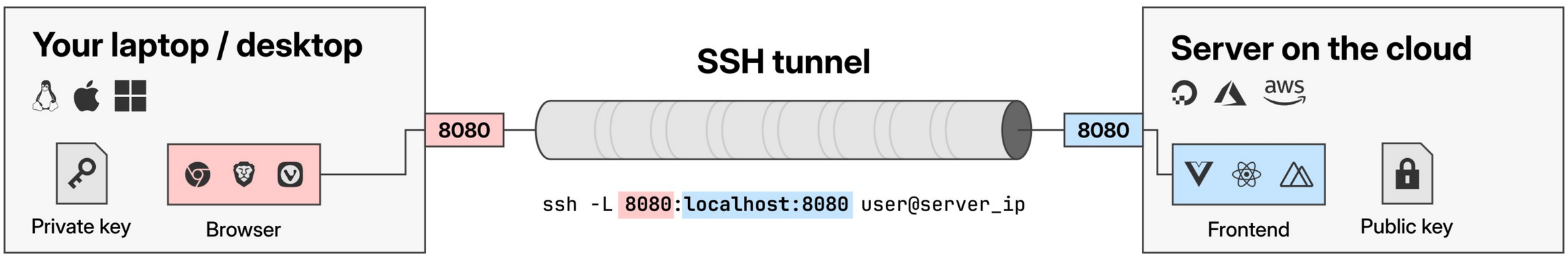
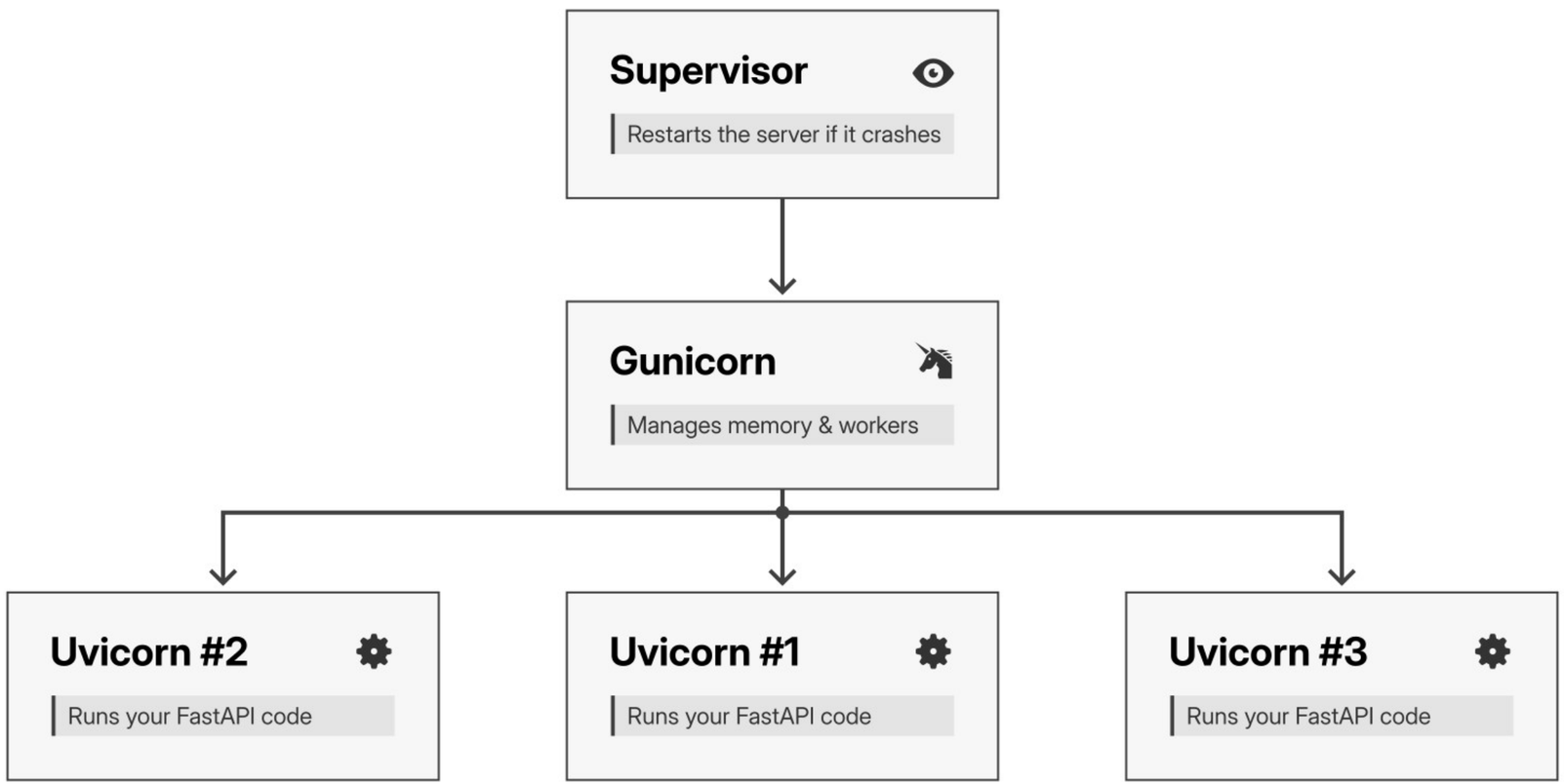
SSH tunneling



SSH tunneling



SSH tunneling



Keeping the app alive